

**Multiple Choice:**

1. A golf ball is hit at an angle of 30 degrees above the horizontal with a speed of 44 m/s. How long is it in the air?
A. 2.25 s
B. 3.00 s
C. 4.50 s
D. 6.00 s
2. A girl moves her brother on a sled at a constant velocity by exerting a force F. The coefficient of friction between the sled and the ground is 0.05. The sled and the rider have a mass of 20 kg. What force is required if she pushed on the sled at an angle of 30 degrees below the horizontal?
A. 11 N
B. 11.4 N
C. 11.7 N
D. 11.9 N
3. A rock is thrown straight upward from the edge of a 30 m cliff, rising 10 m then falling all the way down to the base of the cliff. Find the rock's displacement.
A. 10 m, upward
B. 30 m, upward
C. 40 m, downward
D. 30 m, downward
4. An object with an initial velocity of 4 m/s moves along a straight axis under constant acceleration. Three seconds later, its velocity is 14 m/s. How far did it travel during this time?
A. 13 m
B. 27 m
C. 15 m
D. 31 m
5. A car that's initially traveling at 10 m/s accelerates uniformly for 4 seconds at a rate of 2 m/s^2 in a straight line. How far does the car travel during this time?
A. 26 m
6. The slope of a position versus time graph gives the _____.
A. displacement
B. speed
C. velocity
D. acceleration
7. A rock is dropped off a cliff that's 80 m high. If it strikes the ground with an impact velocity of 40 m/s, what acceleration did it experience during its descent?
A. 0
B. 5 m/s^2
C. 10 m/s^2
D. 8 m/s^2
8. The slope of a velocity versus time graph gives the _____.
A. displacement
B. speed
C. velocity
D. acceleration
9. A rock is dropped from an 80-meter cliff. How long does it take to reach the ground?
A. 2 s
B. 6 s
C. 4 s
D. 10 s
10. One second after being thrown straight down, an object is falling with a speed of 20 m/s. How fast will it be falling 2 seconds later?
A. 10 m/s
B. 30 m/s
C. 20 m/s
D. 40 m/s
11. If an object is thrown straight upward with an initial speed of 8 m/s and takes 3 seconds to strike the ground, from what height was the object thrown?
A. 10 m
12. From a height of 100 m, a ball is thrown horizontally with an initial speed of 15 m/s. How far does it travel horizontally in the first 2 seconds?
A. 10 m
B. 30 m
C. 24 m
D. 44 m
13. A projectile is traveling in a parabolic path for a total of 6 seconds. How does its horizontal velocity 1 s after launch compare to its horizontal velocity 4 s after launch?
A. The horizontal velocity decreased
B. The horizontal velocity increased
C. The horizontal velocity did not change
D. There is not enough information
14. An object is projected upward with a 30° launch angle and an initial speed of 60 m/s. How many seconds will it be in the air?
A. 1 s
B. 4 s
C. 3 s
D. 6 s
15. An object that's moving with constant speed travels once around a circular path. True statements about this motion include which of the following?
I. The displacement is zero.
II. The average speed is zero.
III. The acceleration is zero.
A. I only
B. I and II only
C. I and III only
D. III only
16. Which of the following must always be true?
I. If an object's acceleration is constant, then it must move in a straight line.



- II. If an object's acceleration is zero, then its speed must remain constant.
- III. If an object's speed remains constant, then its acceleration must be zero.
- A. I and II only
B. II only
C. I and III only
D. III only
17. A baseball is thrown straight upward. What is the ball's acceleration at its highest point?
- A. 0
B. 0.5g, downwards
C. g, downwards
D. g, upwards
18. Which one of the following statements is true concerning the motion of an ideal projectile launched at an angle of 45° to the horizontal?
- A. The acceleration vector points opposite to the velocity vector on the way up and in the same direction as the velocity vector on the way down
B. The speed at the top of the trajectory is zero.
C. The horizontal speed decreases on the way up and increases on the way down.
D. The vertical speed decreases on the way up and increases on the way down
19. A stone is thrown horizontally with an initial speed of 30 m/s from a bridge. Find the stone's total speed when it enters the water 4 seconds later. (Ignore air resistance)
- A. 30 m/s
B. 50 m/s
C. 40 m/s
D. 60 m/s
20. A car is traveling in a straight line along a highway at a constant speed of 80 miles per hour for 10 seconds. Find its acceleration.
- A. 0
B. 4 m/s²
C. 2 m/s²
D. 6 m/s²
21. A car is traveling in a straight line along a highway at a speed of 20 m/s. The driver steps on the gas pedal and, 3 seconds later, the car's speed is 32 m/s. Find its average acceleration.
- A. 0
B. 4 m/s²
C. 2 m/s²
D. 6 m/s²
22. The area between the graph and the time axis of a velocity versus time graph gives the ____.
- A. displacement
B. speed
C. velocity
D. acceleration
23. A crate of mass 100 kg is at rest on a horizontal floor. The coefficient of static friction between the crate and the floor is 0.4, and the coefficient of kinetic friction is 0.3. A force F of magnitude 344 N is then applied to the crate, parallel to the floor. Which of the following is true?
- A. The crate will accelerate across the floor at 0.5 m/s².
B. The static friction force, which is the reaction force to F as guaranteed by Newton's third law, will also have a magnitude of 344 N
C. The crate will slide across the floor at a constant speed of 0.5 m/s.
D. The crate will not move.
24. Two crates are stacked on top of each other on a horizontal floor; crate #1 is on the bottom, and crate #2 is on the top. Both crates have the same mass. Compared with the strength of the force F₁ necessary to push only crate #1 at a constant speed across the floor, the strength of the force F₂ necessary to push the stack at the same constant speed
- across the floor is greater than F₁ because
- A. the force of the floor on crate #1 is greater
B. the coefficient of kinetic friction between crate #1 and the floor is greater
C. the force of kinetic friction, but not the normal force, on crate #1 is greater
D. the coefficient of static friction between crate #1 and the floor is greater
25. The coefficient of static friction between a box and a ramp is 0.5. The ramp's incline angle is 30°. If the box is placed at rest on the ramp, the box will
- A. accelerate down the ramp
B. accelerate briefly down the ramp but then slow down and stop
C. move with constant velocity down the ramp
D. not move
26. A crate of mass 20 kg is sliding across a wooden floor. The coefficient of kinetic friction between the crate and the floor is 0.3. If the crate is being pulled by a force of 90 N (parallel to the floor), find the acceleration of the crate.
- A. 1.5 m/s²
B. 4.4 m/s²
C. 2.2 m/s²
D. 5.0 m/s²
27. A crate of mass 100 kg rests on the floor. The coefficient of static friction is 0.4. If a force of 250 N (parallel to the floor) is applied to the crate, what's the magnitude of the force of static friction on the crate?
- A. 100 N
B. 400 N
C. 250 N
D. 500 N



28. An object of mass 5 kg moves at a constant speed of 6 m/s in a circular path of radius 2 m. Find the magnitude of the net force responsible for its motion.
- A. 45 N
B. 90 N
C. 72 N
D. 110 N
29. A 10 kg mass is attached to a string that has a breaking strength of 1,500 N. If the mass is whirled in a horizontal circle of radius 90 cm, what is the maximum speed it can have? (Neglect the effects of gravity)
- A. 8 m/s
B. 15 m/s
C. 12 m/s
D. 18 m/s
30. An object is thrown horizontally with an initial speed of 10 m/s. How far will it drop in 4 seconds?
- A. 80 m
B. 40 m
C. -80 m
D. D. -40 m
31. A block slides down an inclined plane that makes an angle θ with the horizontal. If the coefficient of kinetic friction is μ , find the acceleration of the block.
- A. $g(\sin \theta - \mu g \cos \theta)$
B. $g(\sin \theta + \mu g \cos \theta)$
C. $g(\cos \theta - \mu g \sin \theta)$
D. $g(\cos \theta + \mu g \sin \theta)$
32. A block slides down a frictionless, inclined plane that makes a 30° angle with the horizontal. Find the acceleration of this block.
- A. $g \sin 30^\circ$
B. $2g \sin 30^\circ$
C. $g \cos 30^\circ$
D. $2g \cos 30^\circ$
33. Starting with zero initial angular velocity, a sphere begins to spin with constant angular acceleration about an axis through its center, achieving an angular velocity of 10 rad/s when the angular displacement is 20 rad. What is the value of the sphere's angular acceleration?
- A. 2.5 rad/s^2
B. 4.5 rad/s^2
C. 3.6 rad/s^2
D. 4.9 rad/s^2
34. An object with an initial angular velocity of 1 rad/s rotates with constant angular acceleration. 3 seconds later, its angular velocity is 5 rad/s. Calculate the angular displacement during this time interval.
- A. 6 rad
B. 12 rad
C. 9 rad
D. 15 rad
35. Which of the following is an expression for centripetal acceleration in terms of angular speed (ω)?
- A. $r^2\omega$
B. $r\omega$
C. ω
D. $r\omega^2$
36. The angular velocity of a rotating disk of radius 50 cm increases from 2 rad/s to 5 rad/s in 0.5 s. What is the linear acceleration of a point on the rim of the disk during this time interval?
- A. 3 m/s^2
B. 6 m/s^2
C. 4.5 m/s^2
D. 9 m/s^2
37. A disk of radius 20 cm rotates at a constant angular velocity of 6 rad/s. How fast does a point on the rim of the disk travel in m/s?
- A. 1.2 m/s
B. 1.8 m/s
C. 2.4 m/s
D. 2.8 m/s
38. The angular velocity of a rotating disk of radius 50 cm increases from 2 rad/s to 5 rad/s in 0.5 s. What is the disk's average angular acceleration?
- A. 3 rad/s^2
B. 6 rad/s^2
C. 4.5 rad/s^2
D. 9 rad/s^2
39. A rotating, rigid body makes one complete revolution in 2 s. What is the average angular velocity in rad/s?
- A. 0
B. π
C. 2π
D. 4π
40. The position on an object (measured in meters from the origin, where $x = 0$) moving along a straight line is given as a function of time (measured in seconds) by the equation $x(t) = 4t^2 - 6t - 40$. Find the time at which the object is at the origin.
- A. 1 s
B. 2 s
C. 3 s
D. 4 s
41. The position on an object (measured in meters from the origin, where $x = 0$) moving along a straight line is given as a function of time (measured in seconds) by the equation $x(t) = 4t^2 - 6t - 40$. What is the object's velocity during the time that the object was at the origin?
- A. 22 m/s
B. 14 m/s
C. 26 m/s
D. 32 m/s
42. The outside curve on a highway forms an arc whose radius is 150 ft. If the road is 30 ft wide and its outer edge is 4 ft higher than the inner edge, for what speed is it ideally banked?



- A. 21.1 ft/s
B. 28.9 ft/s
C. 25.4 ft/s
D. 32.3 ft/s
43. A satellite is placed in a circular orbit 100 miles above the earth's surface. Determine the time required for one complete revolution.
- A. 85 min
B. 93 min
C. 90 min
D. 98 min
44. An airplane makes a turn in a horizontal plane without sideslip at 774 kph. At what angle must the plane be banked if the radius of turn is 1.75 km?
- A. 54.1 deg
B. 61.8 deg
C. 58.1 deg
D. 69.6 deg
45. A flywheel 6 ft in diameter accelerates from rest at a constant rate of 4 rpm/sec. Compute the angular velocity on the rim of the flywheel after 10 seconds.
- A. 0.39 rad/s²
B. 0.42 rad/s²
C. 0.35 rad/s²
D. 0.51 rad/s²
46. What is the velocity of a particle after moving by 5 meters if its velocity as a function of position, x , in meters is defined as $v = x^3 - 2x^2 - 5x + 4$?
- A. 44 m/s
B. 54 m/s
C. 48 m/s
D. 66 m/s
47. What is the maximum speed at which a car can make a safe turn into a 60-m radius unbanked curve if the coefficient of static friction is 0.8?
- A. 21.7 m/s
B. 25.4 m/s
C. 23.5 m/s
D. 31.1 m/s
48. A man standing at the foot of a tower throws a ball vertically upward at a certain velocity. On its way up, the time for the ball to reach the top of the tower is 1 sec while on its way down, the time for the ball to reach the top of the tower is 2 sec. Find the height of the tower.
- A. 32.2 s
B. 38.1 s
C. 34.5 s
D. 41.1 s
49. A car starting from rest accelerates to a speed of 120 ft/s in 4 seconds. Assuming a constant acceleration, find the distance traveled by the car in the first 2 seconds.
- A. 54 ft
B. 60 ft
C. 58 ft
D. 64 ft
50. A rocket is released from a jet plane flying horizontally at 1200 kph at an altitude of 2400 m above its target. The rocket thrust gives it a constant horizontal acceleration of 0.6g. Determine the angle between the horizontal and the line of sight to the target.
- A. 12.11 deg
B. 15.23 deg
C. 13.98 deg
D. 18.85 deg

END